Worksheet 25 · Angles & Polygons IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Use angle facts on lines, in triangles and parallel lines • Find interior and exterior angles of polygons • Give reasons for each step in an angle calculation		
Grade: / 54	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Angles & Polygons*

Question 1 (5 marks)

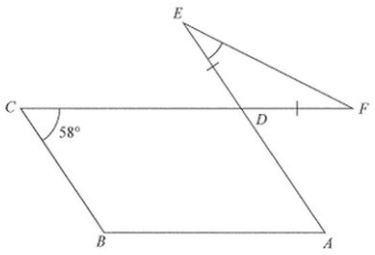


Diagram NOT accurately drawn

The diagram shows a parallelogram *ABCD* and an isosceles triangle *DEF* in which *DE* = *DF*

CDF and *ADE* are straight lines.
Angle *BCD* = 58°

Work out the size of angle *DEF*.
Give a reason for each stage of your working.

.....°

Question 2 (2 marks)

Work out the size of an exterior angle of a regular polygon with 8 sides.

.....°

Question 3 (3 marks)

Each interior angle of a regular polygon is 162°

Work out the number of sides the polygon has.

.....

Question 4 (4 marks)

The diagram shows triangle *ABD*

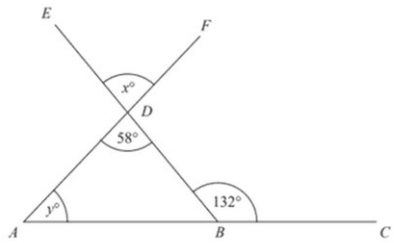


Diagram NOT accurately drawn

ABC, *BDE* and *ADF* are straight lines.

angle *CBD* = 132° angle *ADB* = 58°

(a) (i) Write down the value of *x*

x = ...

(ii) Give a reason for your answer.

.....

(2)

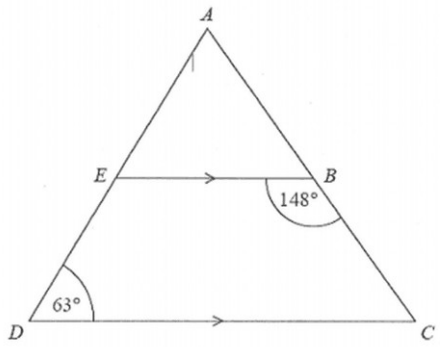
(b) Work out the value of *y*

y = ...

(2)

Question 5 (4 marks)

ADC is a triangle.



AED and *ABC* are straight lines.
EB is parallel to *DC*.

Angle *EBC* = 148°
Angle *ADC* = 63°

Work out the size of angle *EAB*.
You must give a reason for each stage of your working.

Question 6 (4 marks)

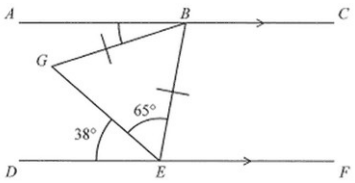


Diagram **NOT** accurately drawn

ABC and *DEF* are parallel lines.

BG = *BE*
Angle *DEG* = 38°
Angle *GEB* = 65°

Find the size of angle *ABG*.

Question 7 (4 marks)

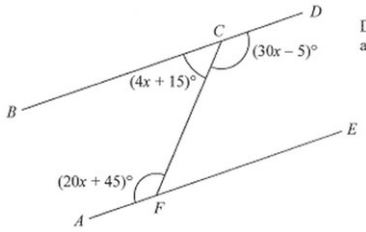


Diagram **NOT** accurately drawn

BCD and *AFE* are straight lines.

Show that *BCD* is parallel to *AFE*.
Give reasons for your working.

Question 8 (4 marks)

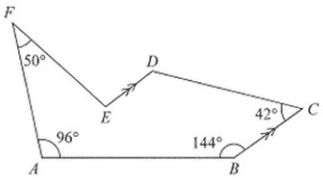


Diagram **NOT** accurately drawn

The diagram shows a hexagon *ABCDEF*.
BC is parallel to *ED*.

Work out the size of the obtuse angle *DEF*.

Question 9 (2 marks)

The diagram shows the isosceles triangle *ABC* in which *AB* = *AC*

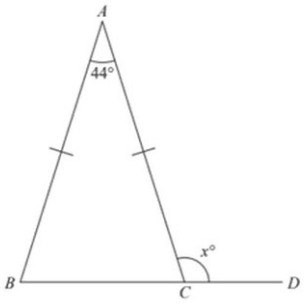


Diagram **NOT** accurately drawn

BCD is a straight line.

Work out the value of *x*.

Question 10 (3 marks)

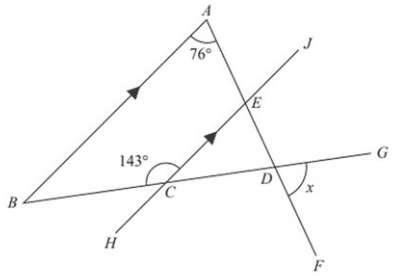


Diagram **NOT** accurately drawn

ABD is a triangle.

AEDF, *BCDG* and *HCEJ* are straight lines.
BA is parallel to *HCEJ*.

Work out the size of the angle marked *x*.

Question 11 (4 marks)

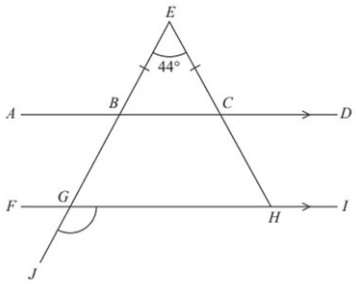


Diagram **NOT** accurately drawn

ABCD and *FGHI* are parallel straight lines.
EBGJ and *ECH* are straight lines.

BE = *CE*
Angle *BEC* = 44°

Work out the size of angle *JGH*.
Give a reason for each stage of your working.

Question 12 (4 marks)

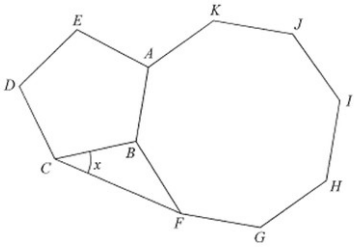


Diagram **NOT** accurately drawn

The diagram shows a regular pentagon, *ABCDE*, a regular octagon, *ABFGHIJK*, and an isosceles triangle, *BCF*.

Work out the size of angle *x*.

Question 13 (4 marks)

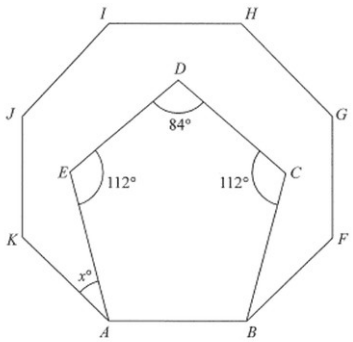


Diagram **NOT** accurately drawn

Pentagon *ABCDE* is drawn inside the regular octagon *ABFGHIJK*.
The pentagon has exactly one line of symmetry.

Work out the value of *x*.

Question 14 (3 marks)

The diagram shows two congruent isosceles triangles and parts of two congruent regular polygons, **X** and **Y**.

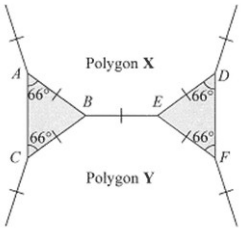


Diagram **NOT** accurately drawn

The two regular polygons each have *n* sides.

Work out the value of *n*.

Question 15 (4 marks)

The diagram shows a regular octagon *ABCDEFGH*.

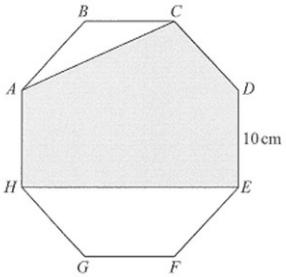


Diagram **NOT** accurately drawn

Each side of the octagon has length 10 cm.

Find the area of the shaded region *ACDEH*.
Give your answer correct to the nearest cm^2